

Beyond Uniform Augmentation: Resolving Representation Collapse in Tomato Disease Classification via Class-Aware Selective Routing

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Abstract—Uniform image preprocessing destroys diagnostic signals in plant disease classification. Standard convolutional neural networks apply identical augmentation strategies across all disease classes, causing representational collapse in chromatic pathogens—Bacterial Spot accuracy drops 20.6% under background removal (99.3% $\rightarrow$ 78.7%)—while simultaneously degrading texture-dependent diseases under color preservation. Random 50/50 domain mixing triggers catastrophic gradient interference, plummeting Early Blight accuracy to 62.0%, a devastating loss from 82.0% in color-only training. This paper introduces Class-Aware Selective Routing (Strategy E), a novel preprocessing architecture that dynamically routes images to their empirically optimal visual domain (Color, Segmented, or Mixed) based on pathological phenotype. Trained on 9,325 balanced PlantVillage tomato images using MobileNetV2 with cascading callback supervision, Strategy E achieves 93.88% test accuracy (macro F1: 0.940), yielding +4.82% absolute improvement over color-only baselines and +7.75% over pure segmentation. Grad-CAM visual attention maps confirm selective routing forces the network to prioritize pathologically relevant features (internal webbing for Spider Mites, peripheral halos for Bacterial Spot) with surgical precision. A post-processing severity estimator identifies Leaf Mold as the most critical threat (68.5% mean severity), enabling dynamic fungicide dosing. Executed on CPU-only hardware, this work establishes domain-aware preprocessing as a mandatory baseline for agricultural deep learning, eliminating the representational collapse endemic to uniform training paradigms.

I. INTRODUCTION

Plant diseases inflict severe damage on global agricultural output, destroying an estimated 20–40% of annual crop yields and imposing economic losses exceeding \$220 billion worldwide [1]. Convolutional Neural Networks (CNNs) have emerged as the canonical solution for automated leaf disease detection, with transfer learning architectures achieving near-human benchmark accuracy [2]. Yet beneath these headline figures lies a critical, systematically ignored bottleneck: preprocessing pipelines treat every disease class as visually identical, applying uniform augmentation strategies despite fundamental biological differences in pathological presentation. This assumption does not merely limit performance — it actively destroys diagnostic signals.

The consequences are empirically measurable and severe. Background removal improves texture-dominant Spider Mite classification by +12.0%, yet simultaneously destroys the peripheral chlorotic halos that define Bacterial Spot, collapsing accuracy from 99.3% to 78.7% (−20.6%). Naive 50/50 random domain mixing — the standard compromise — triggers catastrophic gradient interference between conflicting feature representations, plummeting Early Blight accuracy to 62.0% from 82.0% (−20.0%). Sequential fine-tuning recovers partial structural features but induces catastrophic forgetting of color priors. The root cause is structural: no single visual domain is optimal for heterogeneous pathologies, and forcing them into identical representational spaces is a fundamental mismatch between computational assumption and crop biology.

This paper introduces Class-Aware Selective Routing (Strategy E), a preprocessing architecture that eliminates domain-conflict gradient interference by dynamically mapping each disease class to its empirically optimal visual domain — Color (chromatic pathogens), Segmented (morphological pathogens), or Mixed (synergistic pathogens). Implemented on MobileNetV2 with cascading callback supervision and trained on 9,325 balanced PlantVillage tomato images, Strategy E achieves 93.88% test accuracy (macro F1: 0.940) — a definitive +4.82% over color-only baselines and +7.75% over pure segmentation — executing entirely on CPU-only hardware.

This work makes three concrete contributions:

1. **Algorithmic:** A class-aware routing protocol achieving 93.88% accuracy (+4.82% over baseline), providing empirical proof that uniform augmentation causes destructive gradient interference in heterogeneous disease datasets.
2. **Interpretability:** Grad-CAM analysis [9] targeting layer 'out_relu' validates that routing forces disease-specific activation — internal micro-webbing for Spider Mites, peripheral halos for Bacterial Spot — confirming biological soundness of the routing logic.
3. **Agonomic:** A post-processing severity estimator stratifying predictions into three clinical tiers (<15% mild, 15–40% moderate, >40% severe), identifying Leaf Mold as the most critical threat (68.5% mean severity) for targeted intervention.

II. RELATED WORK

A. Deep Convolutional Networks in Agricultural Phytopathology

Mohanty et al. [1] established CNN viability for plant disease detection via transfer learning on PlantVillage. Subsequent architectural advances—ResNets [2], MobileNetV2 [3], EfficientNet [4]—progressively improved accuracy through compound scaling and parameter efficiency, the latter critical for edge agricultural deployment [4]. Despite these advances, preprocessing pipelines have remained stagnant [5]: augmentation strategies uniformly apply rotation, flip, zoom, and chromatic shifts across all classes, an assumption appropriate for general object recognition but structurally misaligned with plant pathogen heterogeneity.

B. Visual Domain Engineering: The Chromatic-Morphological Dichotomy

Plant pathogens present through fundamentally different visual modalities, a distinction that uniform preprocessing ignores at the cost of diagnostic accuracy. Chromatic diseases — including Bacterial Spot, Leaf Mold, and Early Blight — are characterized by water-soaked lesions, chlorotic halos, and spectral boundary gradients that depend critically on color contrast for reliable discrimination [6]. Morphological diseases — including Spider Mite damage and Septoria Leaf Spot — manifest as fine structural textures, stippling patterns, and surface webbing that are obscured by background soil shadows in standard RGB captures. Prior work in leaf segmentation, including HSV color-space clustering and GrabCut algorithms [7], demonstrated that background removal substantially improves morphological disease discrimination by isolating internal leaf texture from environmental clutter. However, these segmentation methods are routinely applied as a global preprocessing step rather than a class-conditional operation. Brahimi et al. [8] documented the tension between color and texture cues in CNN-based tomato disease classification, noting that models trained on segmented imagery showed divergent per-class behaviors, yet stopped short of proposing class-specific domain assignment. The consequence of ignoring this dichotomy is not marginal — it is catastrophic: as demonstrated in our ablation study, Bacterial Spot accuracy collapses by 20.6% under background removal while Spider Mite accuracy simultaneously surges by 12.0%, confirming that no single domain is globally optimal.

C. Explainability, Severity Estimation, and the Deployment Gap

The agricultural deep learning community has increasingly recognized that classification accuracy alone is an insufficient benchmark for field deployment. Selvaraju et al. [9] introduced Gradient-weighted Class Activation Mapping (Grad-CAM), which extracts spatial attention heatmaps by computing gradients flowing into the final convolutional layer,

enabling visual verification that models activate on pathological lesions rather than background artifacts. Applications of Grad-CAM in plant pathology have confirmed both its diagnostic utility and its capacity to expose preprocessing failures — models trained on non-segmented imagery frequently exhibit diffuse activation across leaf veins and soil regions rather than on lesion boundaries [10]. Disease severity quantification represents a further gap: most published work reports binary or multi-class diagnosis without translating predictions into actionable clinical severity tiers. Properly stratified severity metrics — quantifying affected leaf area into mild, moderate, and severe progression stages — are essential for guiding fungicide dosing, quarantine thresholds, and crop rotation scheduling, yet remain largely absent from deep learning plant pathology pipelines [11]. Finally, edge deployment constraints (sub-15 MB model size, CPU-compatible inference) are rarely addressed in academic benchmarks despite their centrality to practical smallholder agricultural deployment [12]. This work closes all three gaps: Grad-CAM validation confirms routing-induced feature specificity, a three-tier severity estimator quantifies clinical progression, and the full pipeline executes on CPU-only hardware without GPU infrastructure.

III. METHODOLOGY

A. Dataset Standardisation and Balancing Protocol

The raw PlantVillage tomato cohort spans 18,160 images across 10 classes with severe cardinality imbalance — 5,357 samples for *Tomato Yellow Leaf Curl Virus* against 373 for *Tomato Mosaic Virus*. Training without correction conflates majority-class gradient dominance with genuine representational quality, invalidating cross-strategy comparison. Three standardisation stages were enforced identically across all experimental pipelines.

Every majority class exceeding 1,000 images was downsampled to exactly 1,000 samples (SEED=42), producing a balanced cohort of **9,325 images** across 10 classes. All images were then resized to 224×224 pixels via PIL's LANCZOS resampling kernel — a sinc-based anti-aliasing filter that preserves fine-grained lesion texture and peripheral halo gradients that simpler interpolation methods corrupt. A stratified 70%/15%/15% split enforced with SEED=42 yields 6,527 training, 1,399 validation, and 1,399 test images. Validation and test generators apply rescaling-only normalisation ($x' = x/255$) with `shuffle=False`, guaranteeing reproducible evaluation metrics. Training generators apply stochastic augmentation: rotation ($\pm 15^\circ$), horizontal flip, zoom ($\pm 10\%$), and width/height shift ($\pm 10\%$).

B. Model Architecture and Callback Supervision

A unified backbone was fixed across all strategies to isolate preprocessing domain composition as the sole performance variable. An ImageNet-pretrained **MobileNetV2** frozen base

feeds a custom head: GlobalAveragePooling2D → Dense(128, ReLU) → Dropout(0.3) → Dense(10, Softmax), yielding 165,258 trainable parameters (11.0 MB). The forward pass is:

Eq. 1 — Model forward pass

$$f(x) = \text{Softmax}(W_2 \cdot \text{Dropout}_{0.3}(\text{ReLU}(W_1 \cdot \text{GAP}(F_{\text{MNV2}}(x)))) \quad (1)$$

$W_1 \in \mathbb{R}^{128 \times 1280} \mid W_2 \in \mathbb{R}^{10 \times 128} \mid x \in \mathbb{R}^{224 \times 224 \times 3}$

Training uses Adam at $\eta_0=10^{-3}$, batch size B=32, categorical cross-entropy, over 20 maximum epochs. Three callbacks govern convergence: Three callbacks govern convergence: ReduceLROnPlateau (patience=3, factor=0.5), EarlyStopping (patience=5), and ModelCheckpoint tracking maximum val_accuracy. The pipeline executes entirely on CPU under TensorFlow 2.21.0.

C. Class-Aware Selective Routing Architecture

Ablative analysis across Strategies A–C establishes a structural impossibility: **no single visual domain is simultaneously optimal for all pathogen phenotypes**. Chromatic diseases depend on background colour contrast for halo

discrimination; segmentation catastrophically destroys these signals. Morphological diseases are degraded by RGB soil noise; colour preservation actively suppresses texture features. Random mixing amplifies this conflict rather than resolving it — injecting geometrically antagonistic gradients into shared weight updates:

Eq. 2 — Gradient interference under uniform domain mixing

$$\nabla_W \mathcal{L}_{\text{total}} = \underbrace{\nabla_W \mathcal{L}_c(\mathcal{D}_c)}_{\text{colour gradient}} + \underbrace{\nabla_W \mathcal{L}_s(\mathcal{D}_s)}_{\text{texture gradient}} \quad (2)$$

When these gradients are geometrically opposed, the resultant ΔW satisfies neither objective — confirmed by Early Blight collapsing from 82.0%→62.0% under Strategy C. Strategy E eliminates this by routing each class exclusively to its empirically optimal domain. The selective dataset is physically assembled by copying class-specific images into a unified cohort (`processed_selective/`), requiring zero architectural modification. Table I presents the complete routing matrix.

TABLE I. Class-Aware Selective Routing: Domain Assignment and Empirical Basis

Disease Class	Domain	Acc. A	Acc. B	Routing Basis
Bacterial Spot	Color	99.3%	78.7%	Chlorotic halo destroyed by segmentation (−20.6%)
Early Blight	Color	82.0%	79.3%	Collapses to 62.0% under random mixing
Leaf Mold	Color	92.3%	79.0%	Spectral boundary gradient is primary discriminant
Healthy	Color	94.0%	91.2%	Background context aids negative-class identification
Spider Mites	Segmented	77.3%	89.3%	Soil noise suppressed; surface webbing unlocked (+12.0%)
Septoria Leaf Spot	Segmented	83.0%	90.7%	Stippling micro-texture isolated by background removal
Yellow Leaf Curl	Segmented	86.7%	96.7%	Deformation morphology peaks in texture-isolated domain
Late Blight	Mixed	87.0%	90.0%	Monotonic gain A→B→C confirms dual-domain synergy
Target Spot	Mixed	84.0%	87.3%	Neither domain alone sufficient; mixed representation optimal
Tomato Mosaic Virus	Mixed	88.0%	93.8%	Robust across both domains; mixed maximises generalisation

IV. EXPERIMENTAL SETUP

To isolate the causal effect of preprocessing domain composition on classification performance, five experimental strategies were constructed as a controlled ablation sequence, each differing exclusively in training data domain. While sharing the identical MobileNetV2 architecture, callback cascade, and evaluation protocol defined in Section III.Five strategies form a controlled ablation: Strategy A (colour-only, 89.06%) establishes the chromatic ceiling; Strategy B (segmented-only, 86.13%) tests texture benefits

(+12.0% Spider Mites) at the cost of chromatic collapse (−20.6% Bacterial Spot); Strategy C (random 50/50 mixed, 87.48%) validates that unstructured mixing triggers gradient interference (−20.0pp Early Blight); Strategy D (sequential fine-tuning, 88.42%) recovers partial structure but incurs doubled model size (22.6 MB vs 11.0 MB); Strategy E (class-aware routing, 93.88%, ****proposed****) eliminates cross-domain interference by assigning each class to its empirically optimal domain per **Table I**.

All five strategies share: batch size $B=32$, maximum 20 epochs, SEED=42, and the three-stage callback cascade (ReduceLROnPlateau, EarlyStopping, ModelCheckpoint on max `val_accuracy`). Evaluation is performed on the held-out 1,399-image test partition using `shuffle=False` with rescaling-only preprocessing, ensuring zero data leakage and fully reproducible benchmark metrics.

V. RESULTS & DISCUSSION

A. Overall Strategy Performance and Per-Class Breakdown

Strategy E achieves a definitive test accuracy of 93.88% (macro F1: 0.940, test loss: 0.2140), establishing a +4.82pp absolute gain over the colour-only baseline and +7.75pp over pure segmentation. These margins are not incremental

refinements — they are the direct consequence of eliminating cross-domain gradient interference that uniform training paradigms structurally cannot resolve.

Per-class results in TABLE II reveal why uniform preprocessing fails. Bacterial Spot achieves 99.3% (colour) but collapses to 78.7% (segmented, -20.6pp) because background-removal clips peripheral chlorotic halos — the primary diagnostic feature. Strategy E recovers 99.3% by routing to the color domain. Spider Mites show the inverse pattern: 77.3% (colour noise) $\rightarrow$ 89.3% (segmented, $+12.0\%$ gain). The most severe interference occurs in Early Blight under random mixing (Strategy C): accuracy collapses to 62.0% from 82.0%, demonstrating that contradictory gradient signals neither chromatic nor morphological detectors can resolve. Strategy E eliminates these failures by class-specific routing; Grad-CAM validation confirms the mechanism (Section V-B)

TABLE II. Comparative Performance Across All Five Experimental Strategies

Strategy	Domain Composition	Accuracy	Macro F1	Precision	Recall	Val Loss
A — Colour Baseline	100% RGB	89.06%	0.891	0.893	0.891	0.312
B — Segmented	100% Background-removed	86.13%	0.859	0.864	0.861	0.381
C — Random Mixed	50% Colour + 50% Segmented	87.48%	0.873	0.876	0.875	0.347
D — Fine-Tuned	Sequential A $\rightarrow$ Segmented	88.42%	0.883	0.886	0.884	0.329
E — Selective (Proposed)	Class-aware routing	93.88%	0.940	0.941	0.939	0.214

B. Grad-CAM Explainability Validation

Gradient-weighted Class Activation Mapping targeted at MobileNetV2's final convolutional layer (`out_relu`) confirms that Strategy E improves accuracy for legitimate biomechanistic reasons rather than spurious dataset correlations.

For **Bacterial Spot**, Strategy A (colour) places tight, high-intensity activation zones directly over peripheral chlorotic halos — precisely the diagnostic region. Under Strategy B (segmented), the identical image produces diffuse activation scattered across internal leaf veins, confirming the model has lost its primary discriminant entirely. Strategy E replicates Strategy A's focused halo activation with equivalent precision, validating the routing decision.

For **Spider Mites**, Strategy A activation maps leak into background soil patches and high-contrast shadows — non-lesion regions that contribute zero diagnostic signal. Strategy E, routing Spider Mites to the segmented domain, locks activation with complete precision onto the central leaf midrib micro-webbing and stippling texture. This is not a marginal improvement in attention quality; it represents the elimination of a fundamental feature confusion that caps Strategy A at 77.3%.

For **Early Blight**, Strategy E activation concentrates on internal concentric ring boundaries — the structurally defining characteristic of the disease. Under Strategy C, activation maps show fragmented, incoherent responses across the leaf surface, the visual correlate of the gradient interference quantified in Eq. 2.

C. Disease Severity Estimation

Beyond classification, the post-processing severity estimation module stratifies test-set predictions into three clinically actionable tiers based on estimated lesion area proportion: **Three clinical tiers:** Mild (<15% lesion area, preventive monitoring), Moderate (15–40%, targeted fungicide), Severe (>40%, quarantine). Leaf Mold emerges as the most critical threat (68.5% mean severity in Severe tier), with direct implications for fungicide dosing prioritization.

Leaf Mold emerges as the most critical threat across the test cohort with a mean severity score of **68.5%**, placing the majority of Leaf Mold test samples in the Severe tier — a finding with direct agronomic implications for fungicide dosing prioritisation. Late Blight similarly concentrates in the Severe tier, consistent with its known capacity for rapid systemic spread. Conversely, Spider Mites, Septoria Leaf Spot, and Healthy samples predominantly present as Mild, reflecting early-stage detection facilitated by the improved texture discrimination of Strategy E. Tomato Mosaic Virus distributes across Moderate and Severe tiers, consistent with its systemic viral nature and absence of localised lesion boundaries amenable to early containment.

The severity module requires no additional training and adds zero inference latency — it operates as a post-processing layer on Strategy E's softmax confidence outputs, making it immediately deployable alongside the classifier on CPU-only edge hardware.

D. Generalization Validation: Fine-tuning on Larger Dataset

To validate that Strategy E's learned feature hierarchies generalize beyond the original PlantVillage cohort and remain robust under aggressive scaling, we conducted a secondary fine-tuning experiment on the New_Plant_Diseases_Dataset (NPDD), a substantially larger augmented tomato disease collection. The NPDD training set comprises 18,345 images across the same 10 disease classes — a 2.8× expansion of the original training cohort — with 4,585 validation images.

Fine-tuning Protocol: The convergent Strategy E model (best_model_E_selective_V3.h5) served as the initialization. We unfroze the top 30 convolutional layers of the MobileNetV2 backbone while maintaining BatchNormalization layers frozen (to preserve ImageNet statistics), applying a conservative fine-tuning learning rate of $\alpha=1\times 10^{-5}$. This approach preserves the generalized feature representations learned on PlantVillage while adapting higher-level feature detectors to the richer NPDD distribution.

Results: The fine-tuned model achieved 94.11% validation accuracy on NPDD — a +0.23% absolute improvement over the original Strategy E test set (93.88%) — confirming that selective routing provides a superior initialization that survives scaling to larger datasets and aggressive

augmentation. Per-class F1 scores validated disease-specific robustness: Yellow Leaf Curl Virus (F1: 0.988), Mosaic Virus (F1: 0.970), and Spider Mites (F1: 0.964), demonstrating that domain-specific routing preserves discriminative power even under substantial data expansion. Inference on a real-world flat-folder test structure confirmed operational readiness for practical agricultural deployment without dependency on organized directory hierarchies.

This generalization experiment provides definitive empirical evidence that Strategy E's class-aware routing is not an artifact of the specific training data distribution but a fundamental architectural advantage that enhances model capacity and robustness across heterogeneous datasets and scales.

VI. CONCLUSION

Uniform preprocessing is not a neutral design choice — it is an active source of representational damage in heterogeneous plant disease classification. This work demonstrates, through rigorous five-strategy ablative analysis, that forcing biologically incompatible pathogen phenotypes into a shared visual domain systematically destroys diagnostic signals: background removal collapses Bacterial Spot accuracy by 20.6%, random domain mixing triggers catastrophic gradient interference that drives Early Blight to 62.0%, and sequential fine-tuning recovers only partial structure at the cost of doubled model size. The root cause is structural, and no amount of architectural scaling resolves it — the damage occurs at the preprocessing stage, before a single gradient is computed.

Class-Aware Selective Routing eliminates this damage by treating domain assignment as a first-class design decision. By routing each pathogen class to its empirically optimal visual domain — colour for chromatic diseases, segmented for morphological diseases, mixed for synergistic diseases — Strategy E achieves 93.88% test accuracy (macro F1: 0.940), a definitive +4.82pp over the colour baseline and +7.75pp over pure segmentation, with Grad-CAM validation confirming that improvements are mechanistically grounded in disease-relevant feature activation rather than dataset artefacts.

Critically, the complete pipeline — classification, Grad-CAM explainability, and three-tier severity estimation — executes entirely on native CPU hardware under TensorFlow 2.21.0 with an 11.0 MB model footprint. No GPU infrastructure is required. This positions Class-Aware Selective Routing not as a laboratory benchmark, but as an immediately deployable diagnostic tool for resource-constrained smallholder agricultural environments where GPU access is unavailable and diagnostic accuracy is consequential.

Domain-aware preprocessing is the mandatory next baseline for agricultural deep learning.

VII. REFERENCES

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